Candida Demorla

Linux Admin - Lab 1

Server Setup



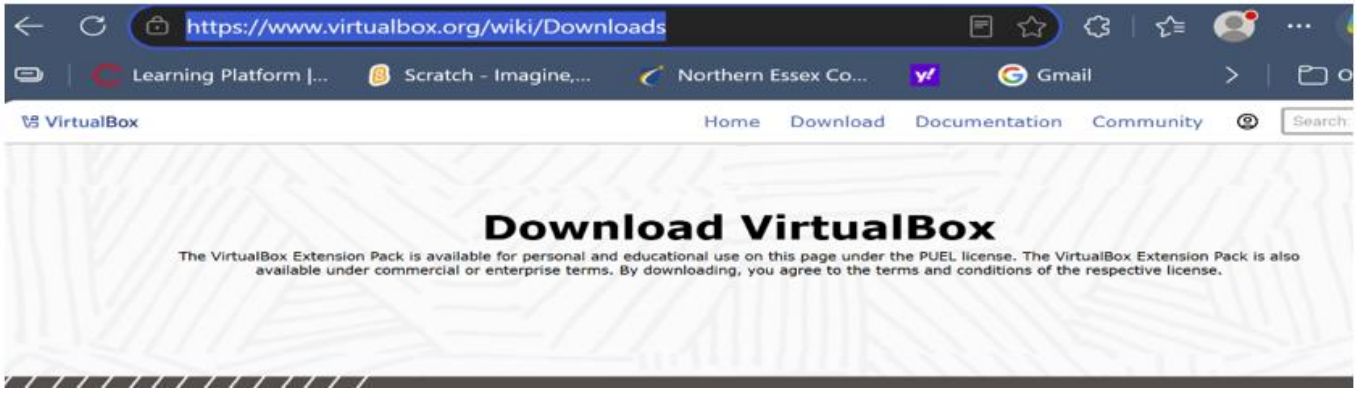
Introduction:

In this lab the goal is to install multiple linux servers on a Virtual Machine or VM. A Virtual Machine is a virtual machine running inside a host machine or Computer. The virtual machine also known as guess machine has its own dedicated cpt, ram and disk space and you can install multiple VM on the Host machine. The Virtual Machine used in this lab is the **Oracle VirtualBox** because it was previusly intalled and updated on the host machine. The **KVM**(Kernel-based Virtual Machine) used in linux enviroments is also an option.

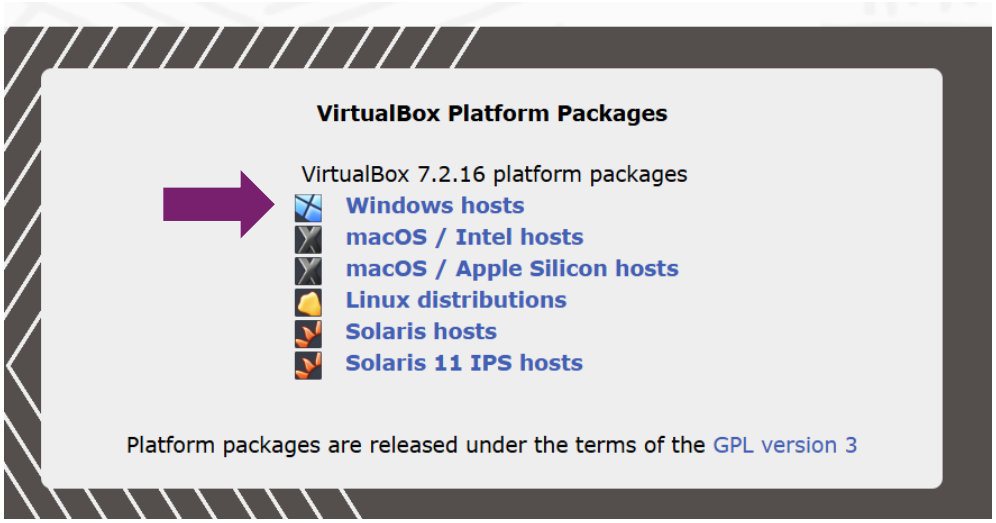
This document focust on downloading steps by steps descriptions on downloading and intalling the CentOS server without the GUI and the Obuntu server. The steps also cover in detail how to set up internet conectivity and share files between servers.

Create a VM

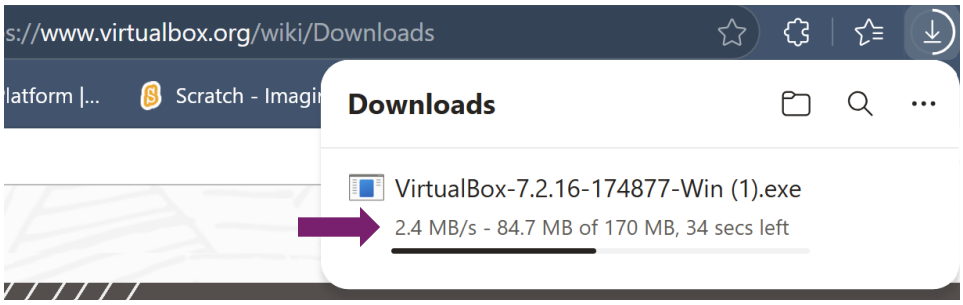
Step 1 - Download VirtualBox: https://www.virtualbox.org/wiki/Downloads



Step 2 - Choose the: VirtualBox 7.2.16 platform packet for the windows host computer.

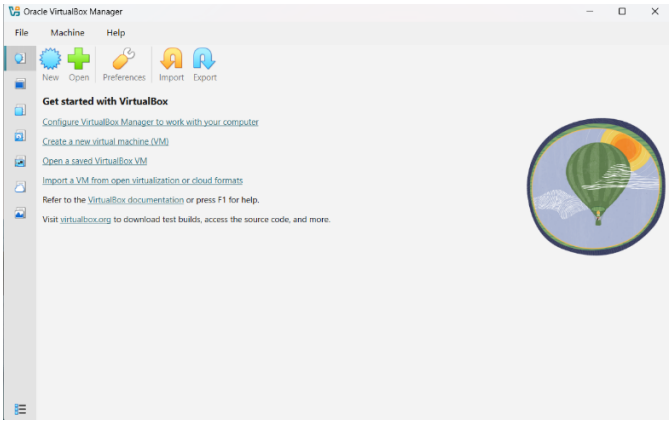


Step 3 – Verify the packet: Once selected you can confirm that the downloading packet is the correct one by checking the filename on the download status



Step 4 - How to install Select: The VirtualBox installation located in the host pc downloads folder and proceed with the installation process

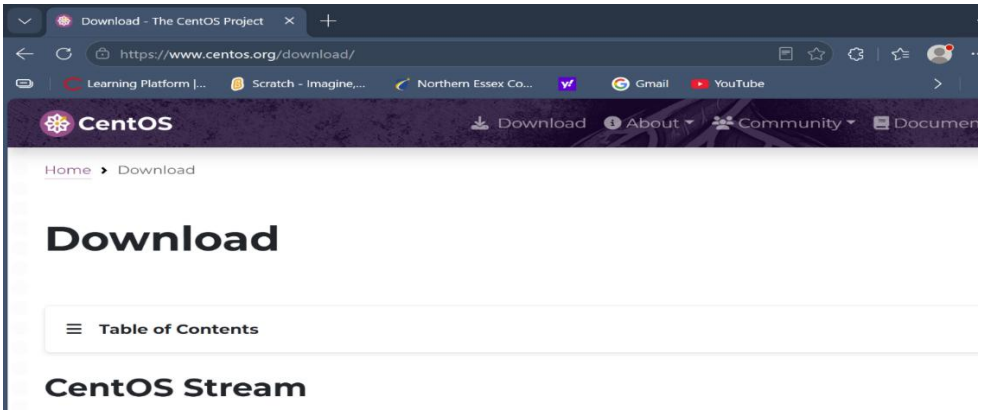
Once installed the **VirtualBox main screen** should look like this image and after that you can proceed to download and install the servers



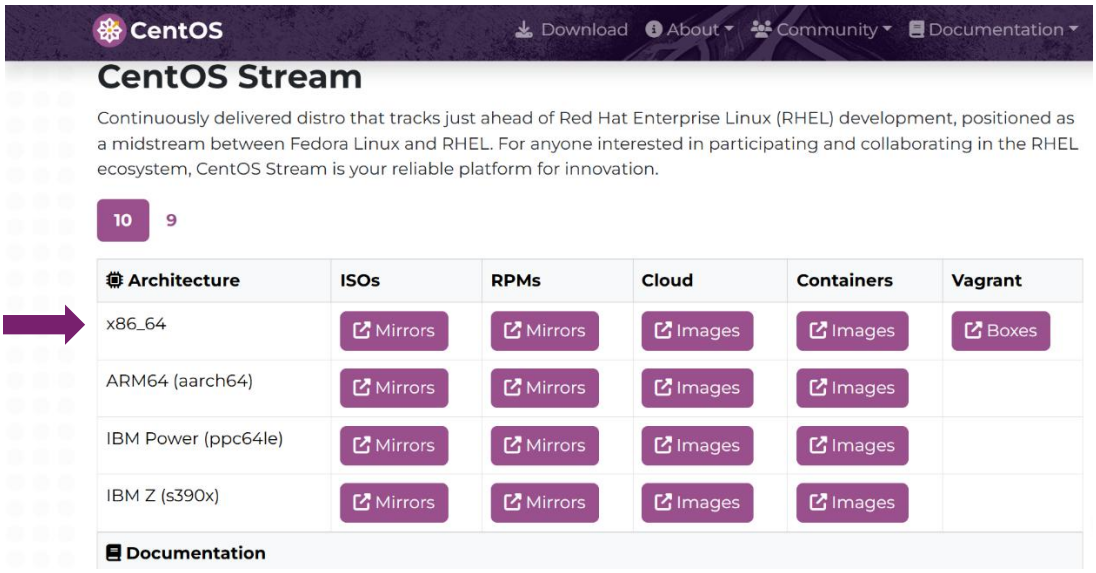
Centos:

Step 1 - Download Centos:

https://www.centos.org/download/



Step 2 - Choose the Centos Architecture: X86_64

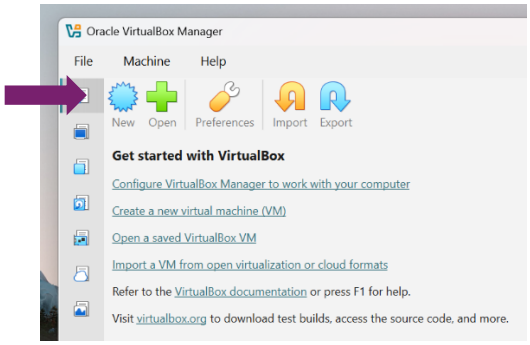


Step 3- Verify the Architecture packet: Once selected you can confirm that the downloading Architecture is the correct one by checking the filename on the download status

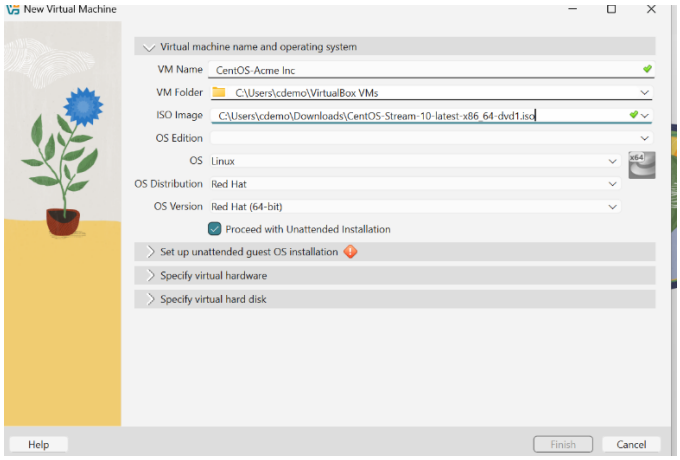
Step 4 - How to create the CentOS VM:

1. VirtualBox Main Screen click the new blue button on top
2. Complete the VM Naming and select the CentOS ISO image on VirtualBox from the host machine downloads folder and make sure to uncheck the Proceed with Unattended Installation
3. Specify the virtual hardware this will include the base memory and the number of CPUs needed – on this server will have base memory: 4096 MB, and 2 CPUs
4. Specify the size of the virtual hard disk needed for your VM and click Finish – this server will have 25GB hard disk space

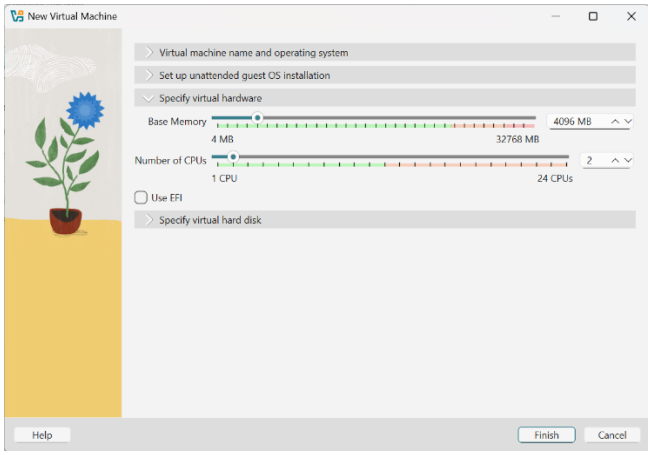
VirtualBox Create New Machine Image



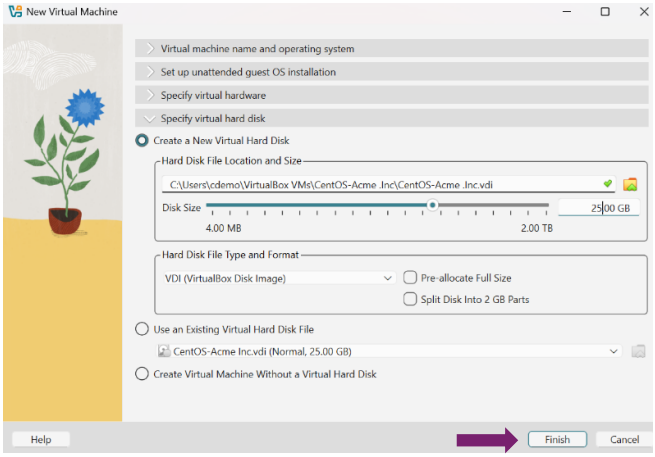
CentOS VM Naming and ISO image initial setup



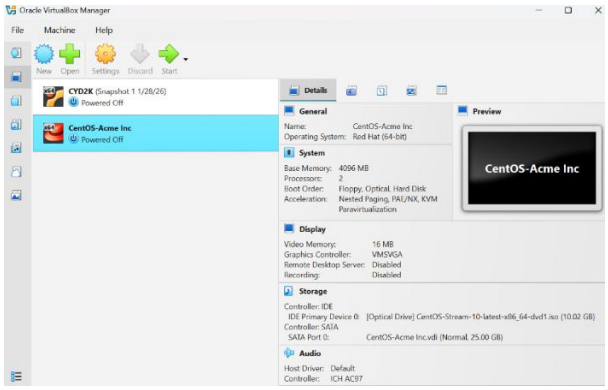
Base memory and CPUs setup



Virtual hard disk setup and Finish creating the VM



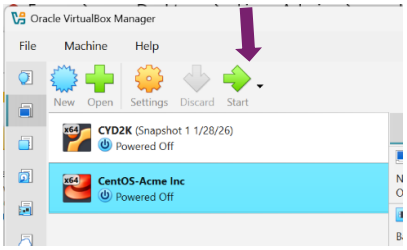
Centos VM Created

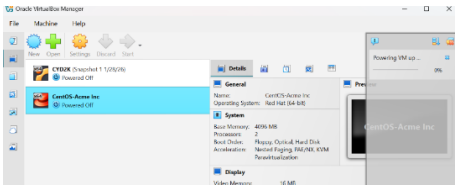


Show how to install OS

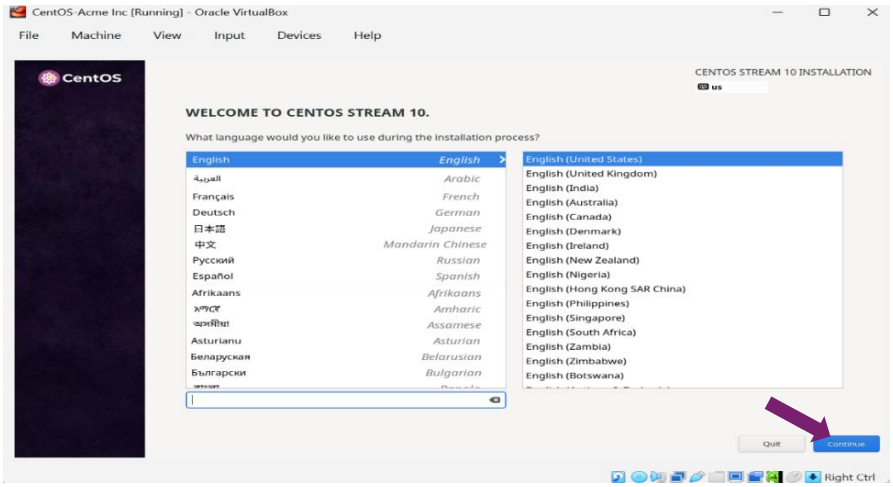
1. CentOS Installation Process – make sure that the CentOS-Acme Inc VM is highlighted and press the green start arrow on the top
2. Select Language screen – after selecting the desired language click continue
3. CentOS Installation Summary Page – Here you need to create the destination, a root password and a user account after that click begin installation
4. Software selection screen – make sure that in the base environment only the Server option is selected as shown and then click Done on top left (The server only option will make sure that your server is installed without the GUI.)
5. When the installation is complete – click **Reboot System**
6. After the server reboot – the terminal will prompt you to login – here you will use the username and password previously created for the CentOS server
7. CentOS first terminal **– checking the network interface details type the command** ip addr
8. Test the internet **- Type** *-c 4 ping google.com*

These images show that the CentOS-Acme Inc VM is highlighted, click the start button and power on

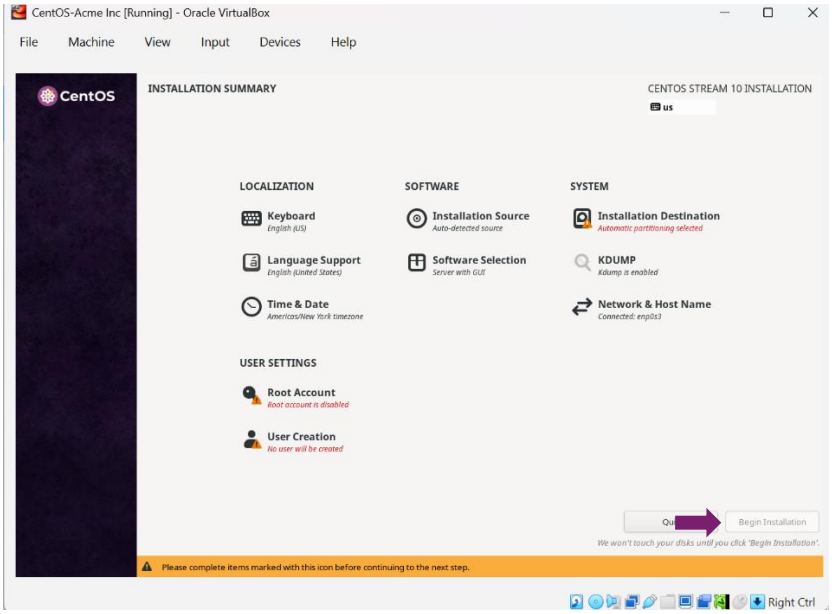




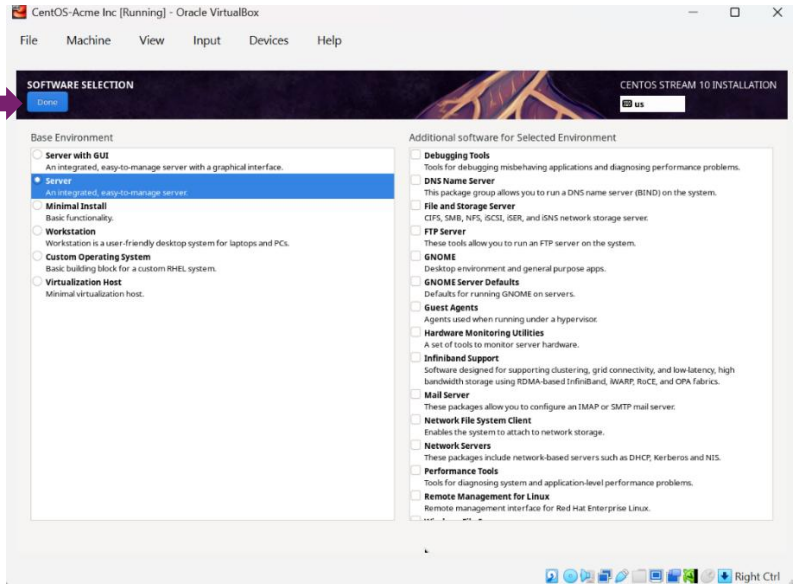
This is the Welcome to CentOS and select Language screen



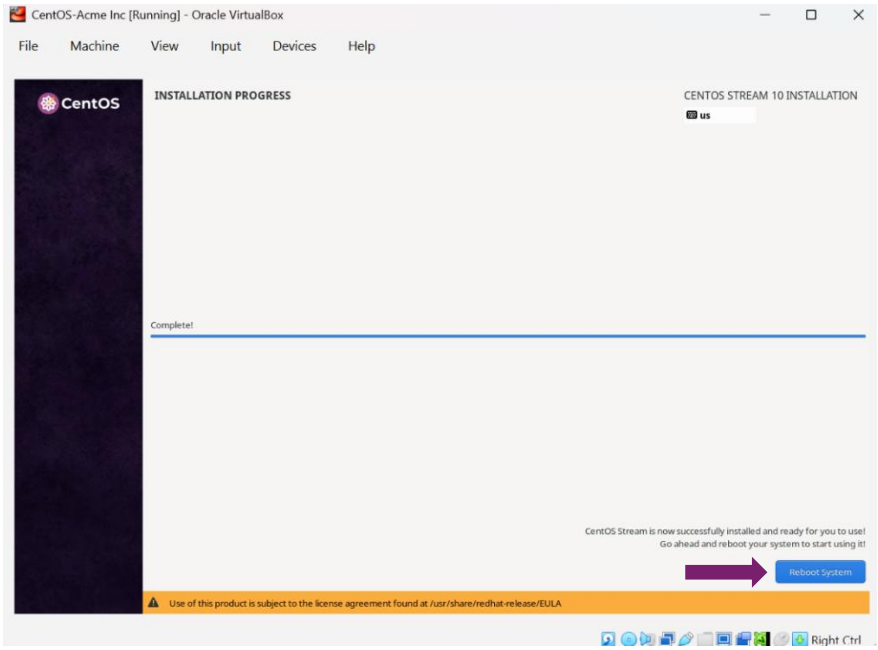
CentOS Installation Summary Page – where you create the destination, a root password and a user account and the begin installation



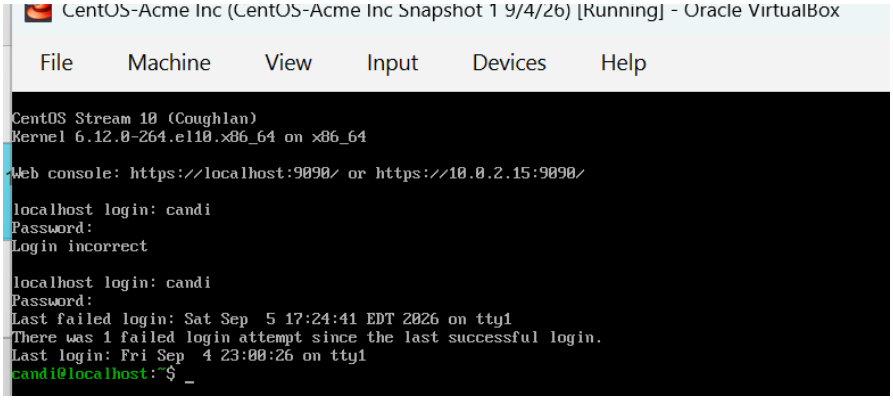
Software selection screen



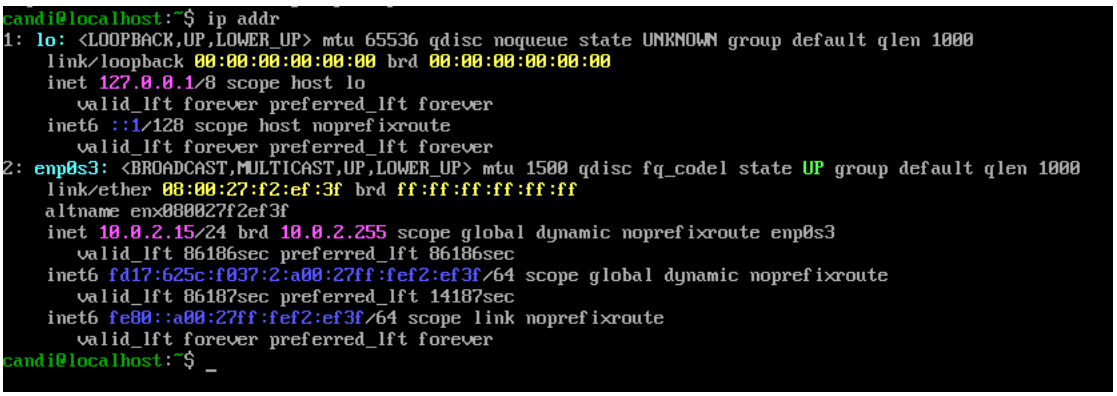
This is the installation complete screen



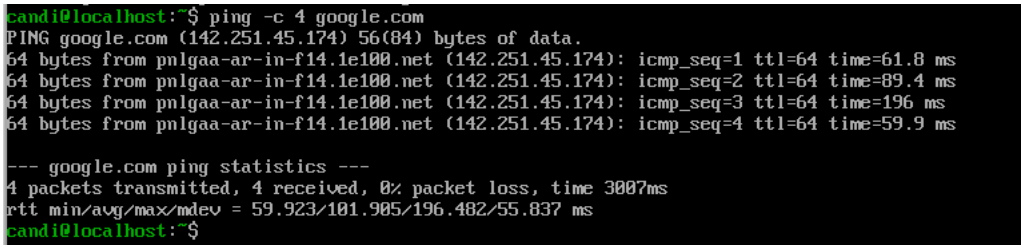
CentOS server successful login



CentOS first terminal use – checking the network interface details by using the command ip addr



Ping google.com

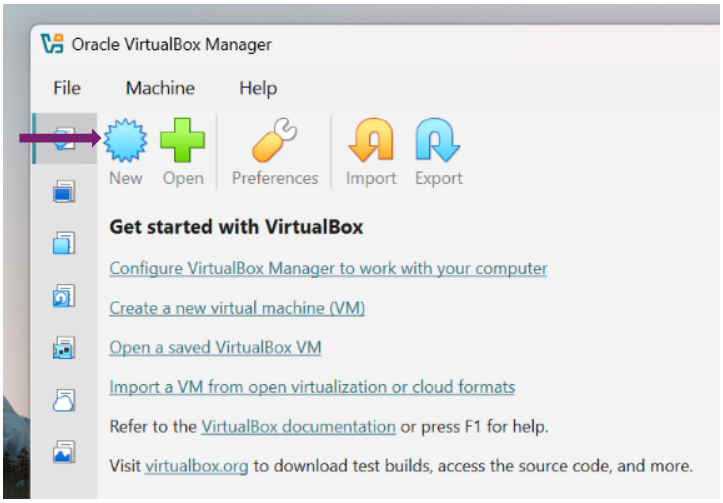


Ubuntu:

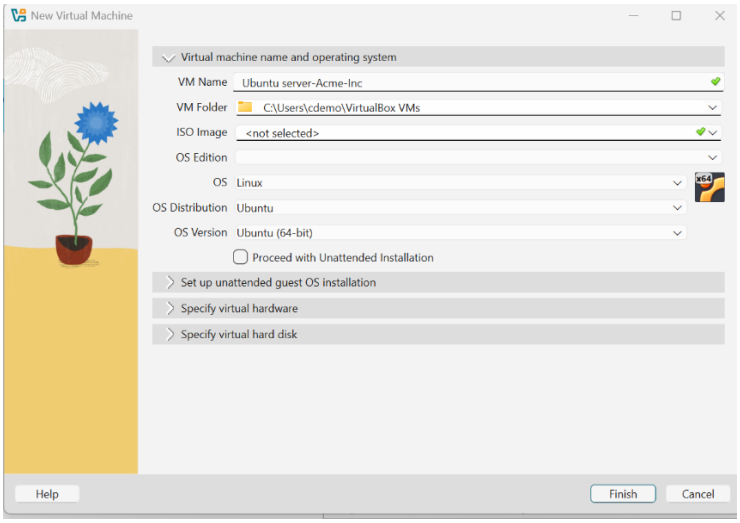
How to create the Ubuntu VM:

1. VirtualBox Main Screen click the new blue button on top
2. Complete the VM Naming and select the Ubuntu ISO image on VirtualBox from the host machine downloads folder and make sure to uncheck the Proceed with Unattended Installation
3. Specify the virtual hardware this will include the base memory and the number of CPUs needed – on this server will have base memory: 4096 MB, and 2 CPUs
4. Specify the size of the virtual hard disk needed for your VM and click Finish – this server will have 25GB hard disk space

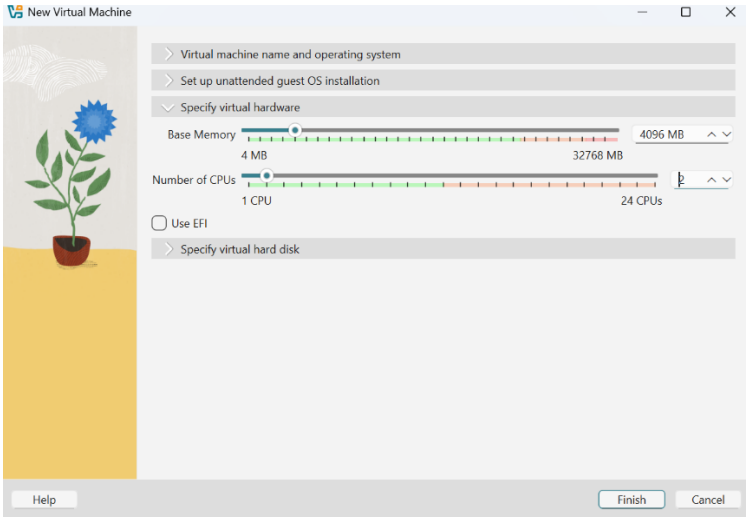
VirtualBox Create New Machine Image



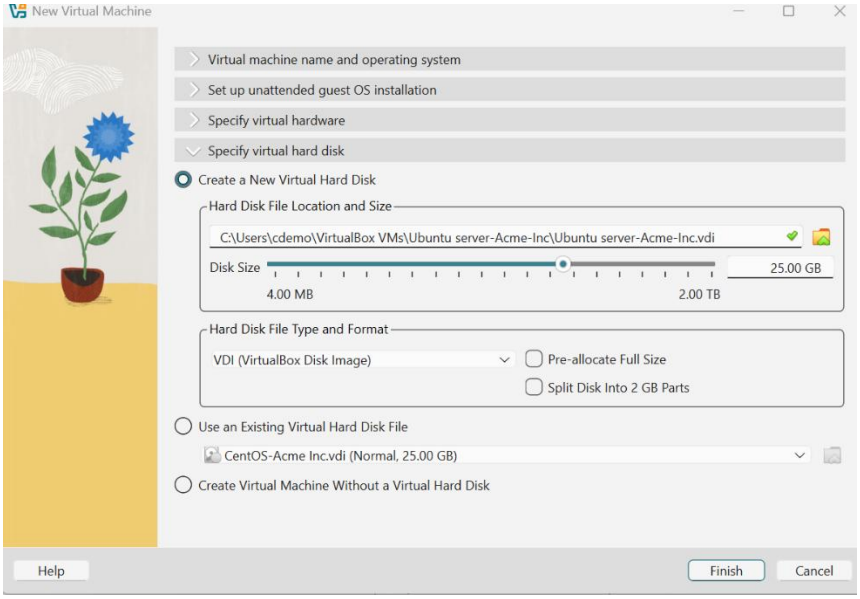
VM Naming and select the Ubuntu ISO image on VirtualBox initial setup



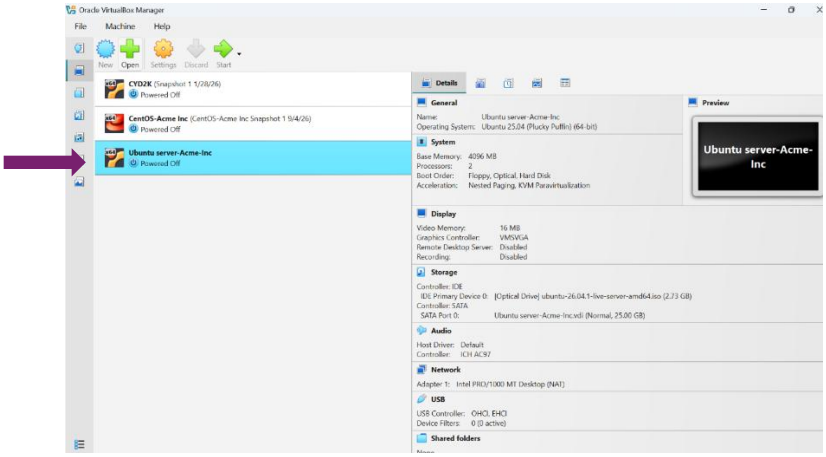
Base memory and CPUs needed setup



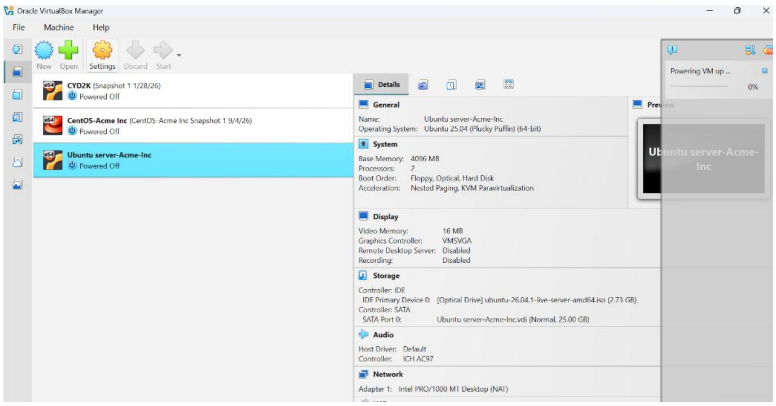
Virtual hard disk setup and click Finish creating the MV



These images show that the Ubuntu server-Acme Inc VM is highlighted, click the start button and power on



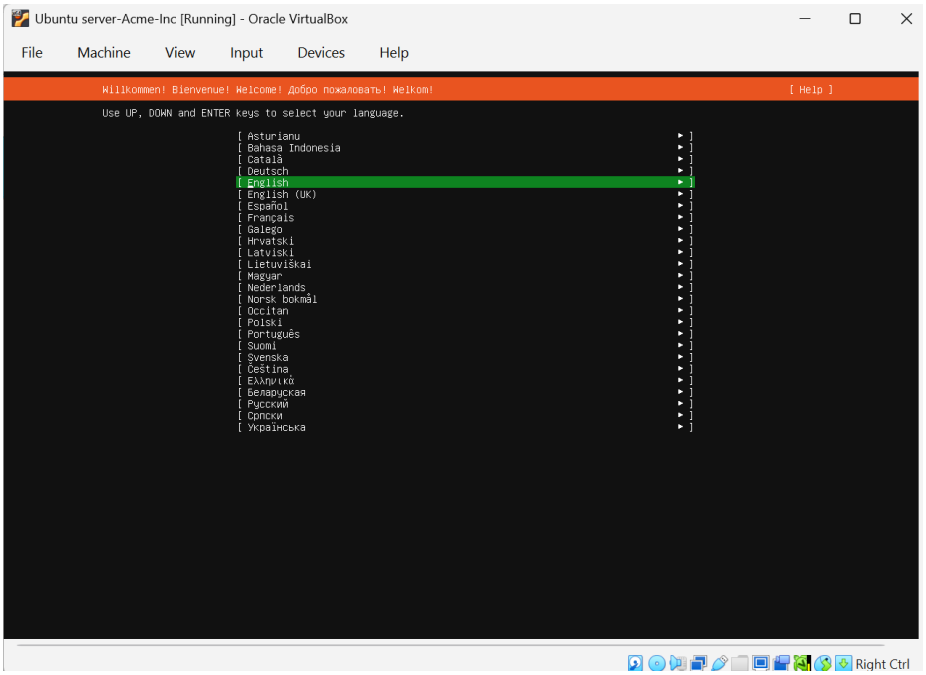
This image shows that the VM is powering on



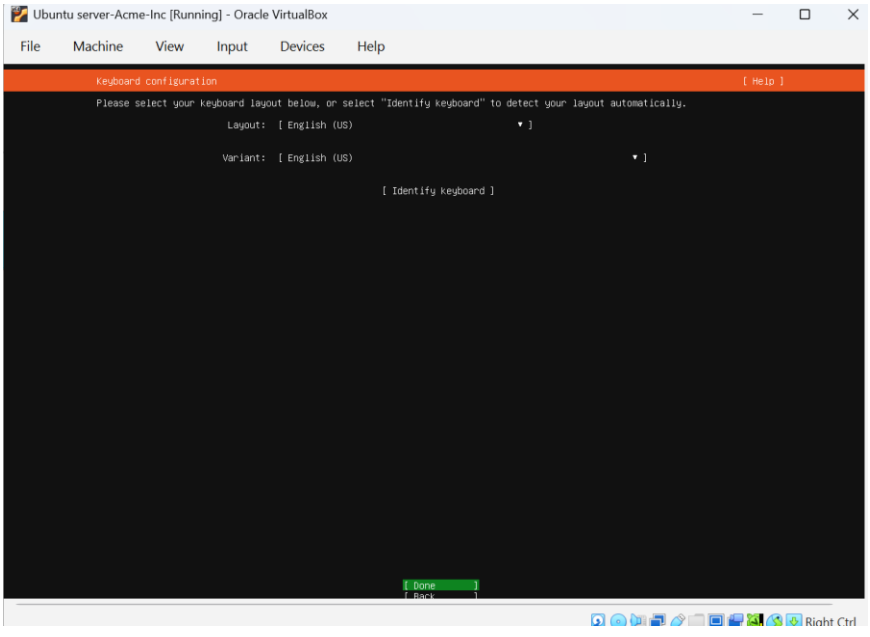
Show how to install OS

1. Ubuntu Installation Process – make sure that the **Ubuntu server-Acme Inc** VM is highlighted and press the green start arrow on the top
2. Select Language screen – after selecting the desired language click continue
3. Select the keyboard configuration settings click done highlighted on green at the bottom and press enter
4. Choose the type of installation in this case the first option server and select done and press enter
5. Network configurations – make sure that a network configuration that looks like this is selected, the values may change depending on your network settings this is just the one shown on the image for this particular installation
   enp0s3
   DHCPv4 10.0.2.15/24
   DHCPv6 fd17:625c:fo37:2:a00:27ff:fe05:b4a2:/64
6. Ubuntu Proxy - select done and press enter
7. Ubuntu Archive mirror configuration - select done and press enter
8. Guided storage configuration – select use entire disk, set up this disk as LMV group and select done and press enter
9. On the storage configuration page do not change anything and select done and press enter
10. Complete the profile configuration by entering your name, your servers name, a username, and a password, confirm the password and select done and press enter
11. On upgrade to ubuntu pro – select skip for now and select done and press enter
12. SSH configuration - select install OpenSSH server and import ssh keys AUTHORIZED KEYS
13. Feature server snaps – select done and continue with installation you can install those if needed later in the terminal
14. Once the Ubuntu installation process is complete select reboot and the machine will restart
15. After restart the remove installation media message will appear on the screen, and you need to click Devices, Optical Drivers and Remove Disk from Virtual Drive
16. After the server reboot – the terminal will prompt you to login – here you will use the username and password previously created for the Ubuntu- server-Acme Inc
17. Ubuntu first terminal **– checking the network interface details type the command** ip addr
18. Test the internet **- Type** *-c 4 ping google.com*

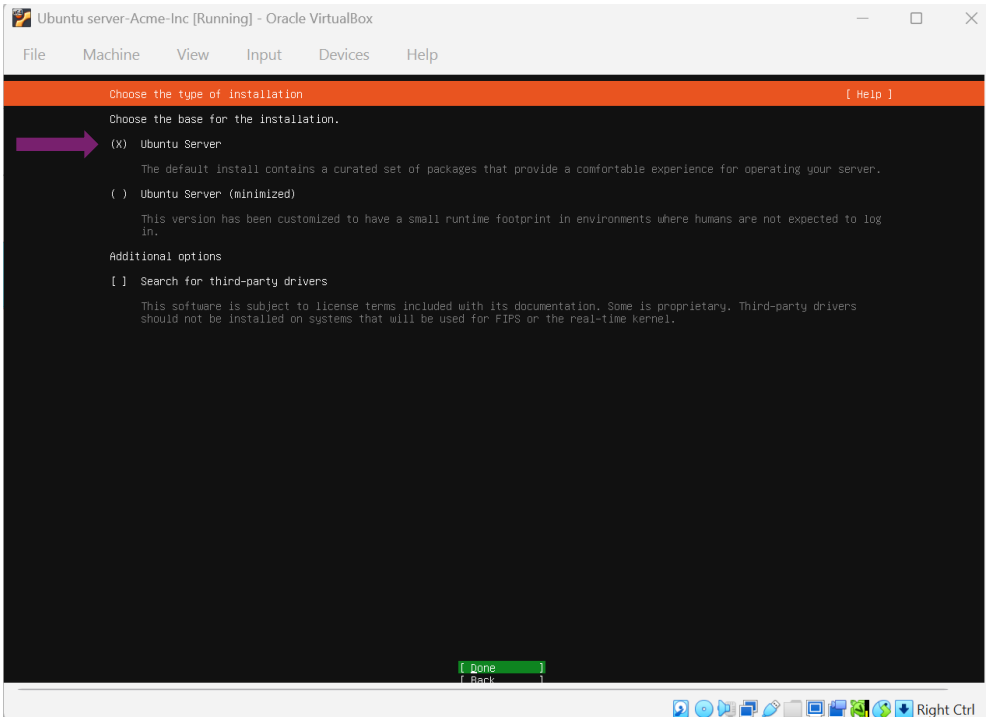
Select Language screen



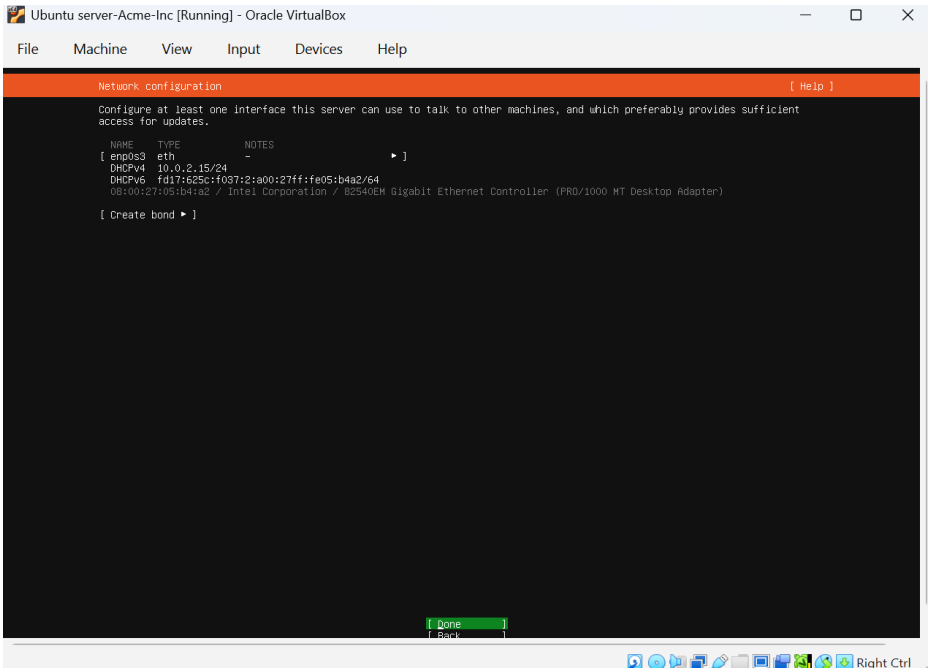
Keyboard configuration settings



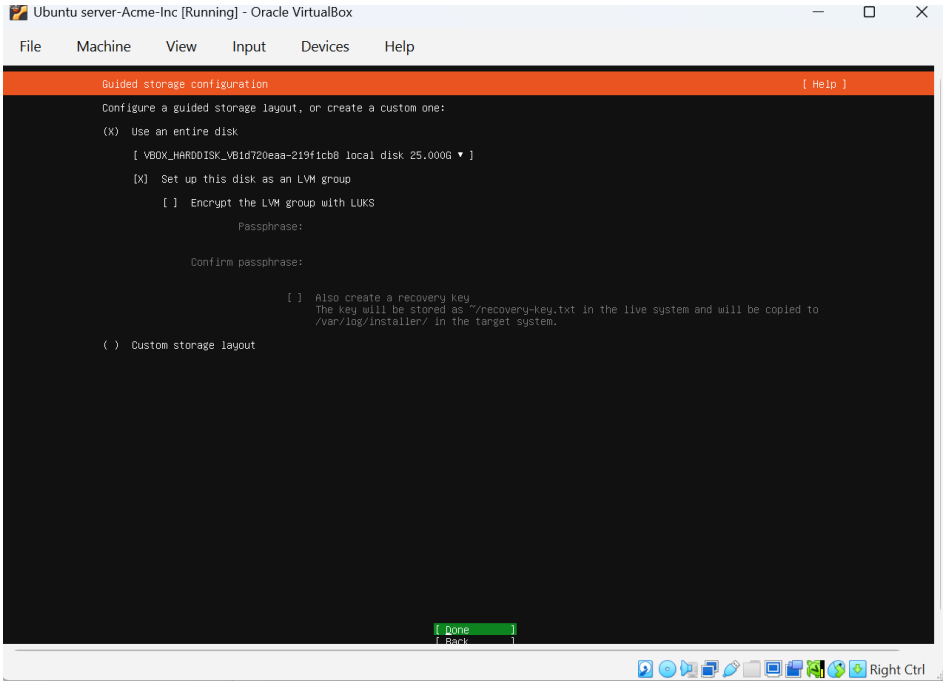
Choose the type of installation – Ubuntu Server



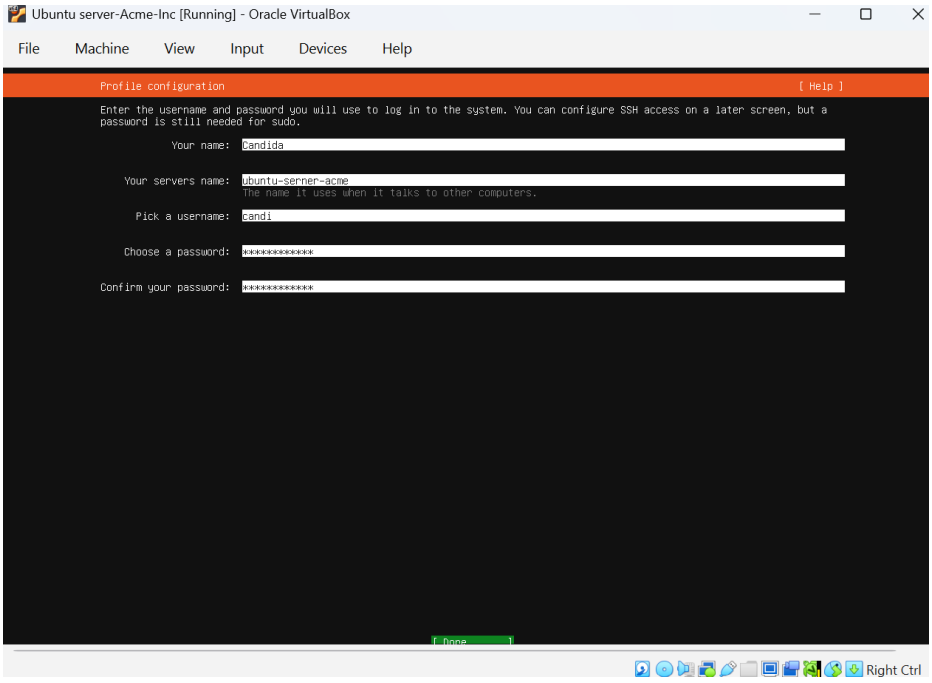
Network configurations – make sure it looks like the image below and select done and press Enter



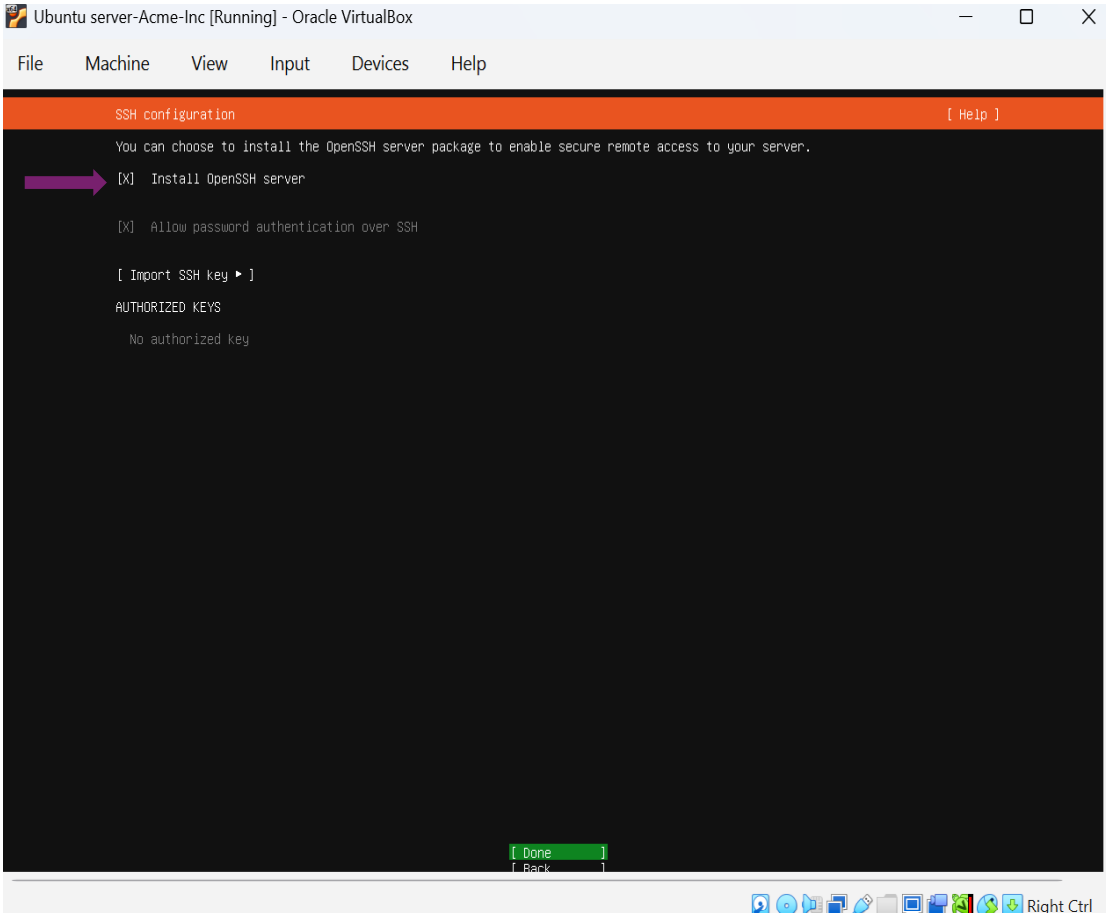
Guided storage configuration



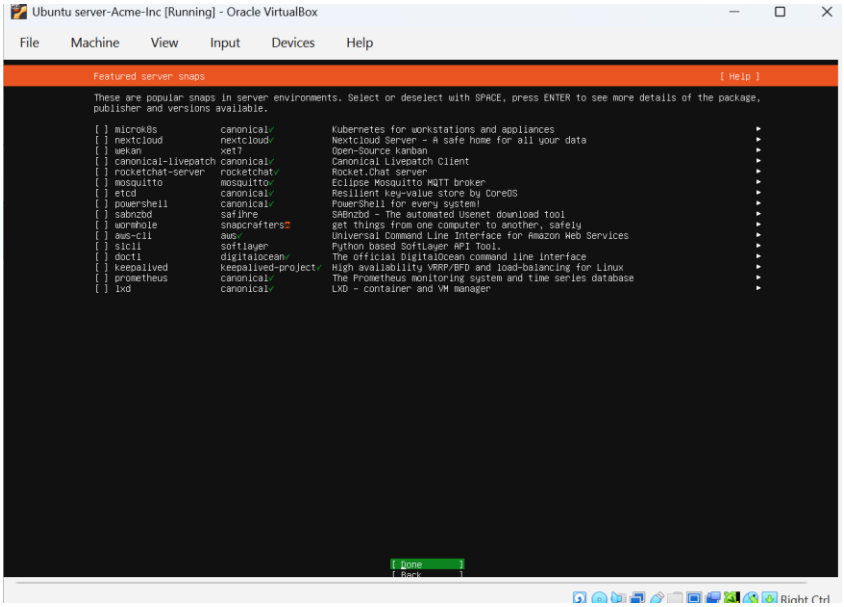
Complete the profile configuration



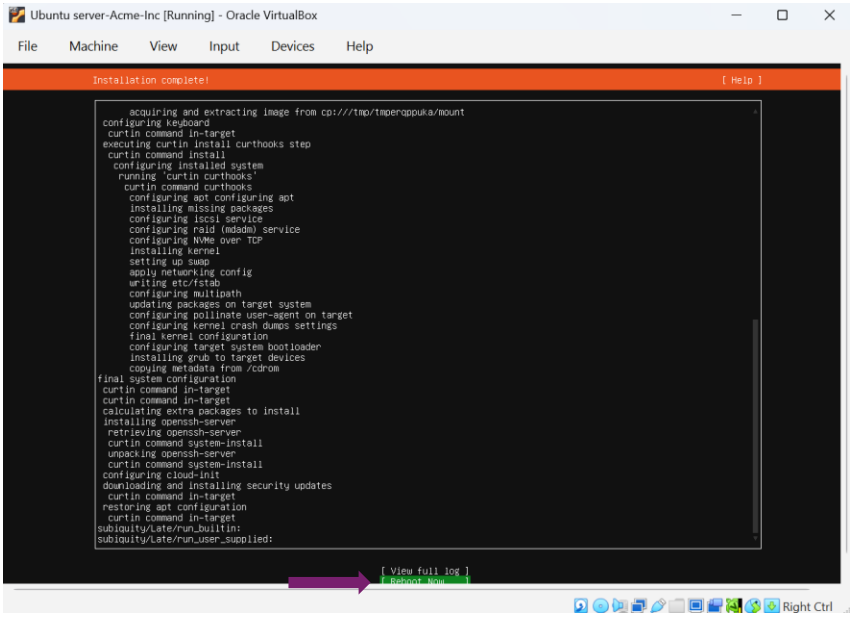
SSH configuration



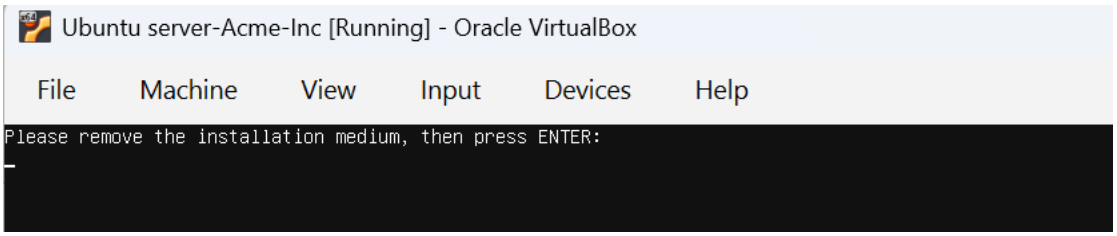
Feature server snaps



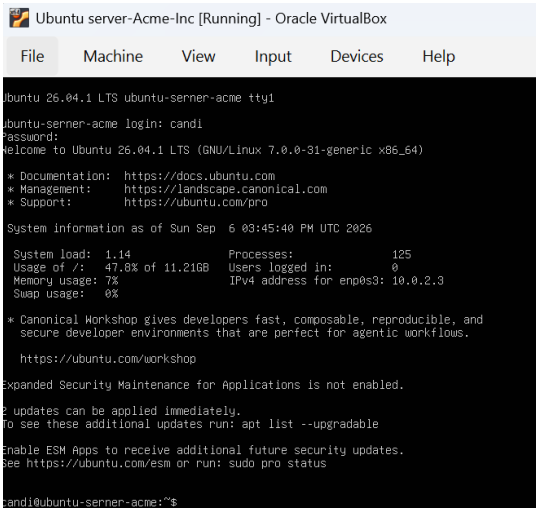
Ubuntu installation process complete



Remove installation media message



Ubuntu successful login



Ubuntu first terminal use – checking the network interface details by using the command *ip addr*

```
ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:05:b4:a2 brd ff:ff:ff:ff:ff:ff
    altname enx08002705b4a2
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 84672sec preferred_lft 84672sec
    inet6 fd17:625c:1057:2:a00:27ff:fe05:b4a2/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86354sec preferred_lft 14354sec
    inet6 fe80::a00:27ff:fe05:b4a2/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
candi@ubuntu-server-acme:~$
```

Ping google.com to test internet

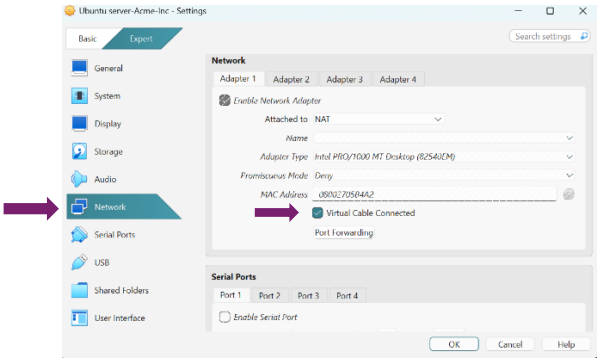
```
candi@ubuntu-server-acme:~$ ping -c 4 google.com
PING google.com (142.250.188.14) 56(84) bytes of data.
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=1 ttl=64 time=19.4 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=2 ttl=64 time=77.0 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=3 ttl=64 time=114 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=4 ttl=64 time=57.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 19.408/66.923/113.916/34.123 ms
candi@ubuntu-server-acme:~$
```

Ping google.com test to show a simulated failure ping and how to troubleshoot

1. **On the ubuntu server running select machines from the top menu**, settings, network adapter 1 – disconnect the virtual cable by unchecking that option and close
2. **In the terminal Type** *ping -c 4 google.com* **a resulted failure ping message will appear**
3. Reconnect the virtual cable and **type in the terminal** *ping -c 4 google.com* and you should get a successful ping.

This test is shown in the pictures below:




```
candi@ubuntu-server-acme:~$ ping -c 4 google.com
ping: google.com: Temporary failure in name resolution
candi@ubuntu-server-acme:~$ ping -c 4 google.com
PING google.com (142.250.188.14) 56(84) bytes of data.
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=1 ttl=64 time=169 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=2 ttl=64 time=77.7 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=3 ttl=64 time=125 ms
64 bytes from nuq04s35-in-f14.1e100.net (142.250.188.14): icmp_seq=4 ttl=64 time=155 ms

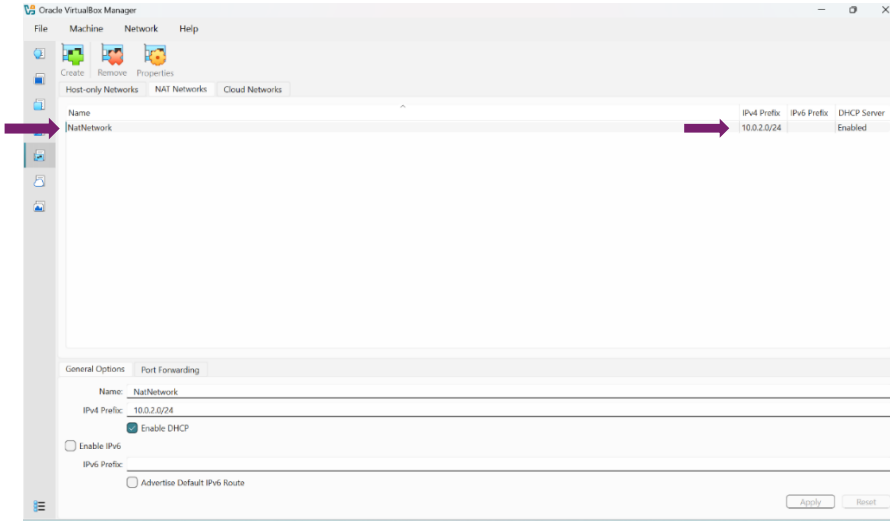
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3555ms
rtt min/avg/max/mdev = 77.735/131.770/168.889/34.935 ms
candi@ubuntu-server-acme:~$
```

How to share a file:

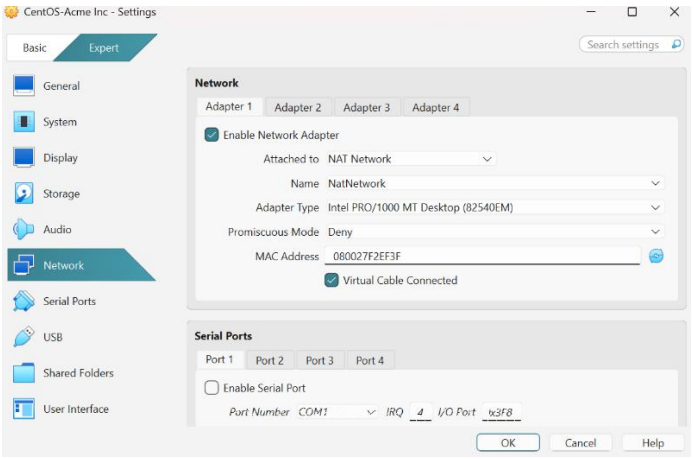
To share a file first create a NAT Network on VirtualBox

1. Power off both **CentOS and Ubuntu Virtual Machines**
2. **On the VirtualBox Menu on top select** File, then Tools and Network
3. Select NAT **Network** and click **create**
4. **On the CentOS server machine, go to network, network adapter 1, attached to NAT Network created on VirtualBox**
5. **On the Ubuntu server machine, go to network, network adapter 1, attached to NAT Network created on VirtualBox**
6. **Once both servers are connected to the same NAT Network the DHCP will assign their own IP address withing the same network which will allow them to communicate with each other and share files**
7. **Make sure that SSH service is running in CentOS – type *systemclt status sshd* and press Enter**
8. **Make sure that SSH service is running in Ubuntu – type *systemclt status ssh* and press Enter**
9. **Test connectivity between servers – From the CentOS terminal, type** ping -c 4 10.0.2.3 **to ping the Ubuntu server. From the Ubuntu terminal, type** ping -c 4 10.0.2.15 **to ping the CentOS server.**
10. **Create a file test file on Ubuntu – type *echo "File Transferred from Ubuntu to CentOS using SCP" > testfile.txt***
11. **Meke sure that the file exists and verify its contents – type *cat testfile.txt* (The message** "File transferred from Ubuntu to CentOS using SCP" **will be shown**
12. **Transfer the testfile.txt to CentOS ip address– type *scp testfile.txt candi@10.0.2.15:~/***
13. **When prompted to continue connecting to the CentOS – type *yes* and press Enter**
14. **When prompted for the password, enter the password of the candi user on the destination CentOS server.**
15. **Check if the file arrived on the CentOS**
16. **type *ls -l ~/test file.txt* to check that the transferred file exists**
17. **Type – *cat ~/testfile.txt* to show the contests of the transferred file.**
18. **The message "File Transferred from Ubuntu to CentOS using SCP" is the confirmation that the file was transferred successfully.**

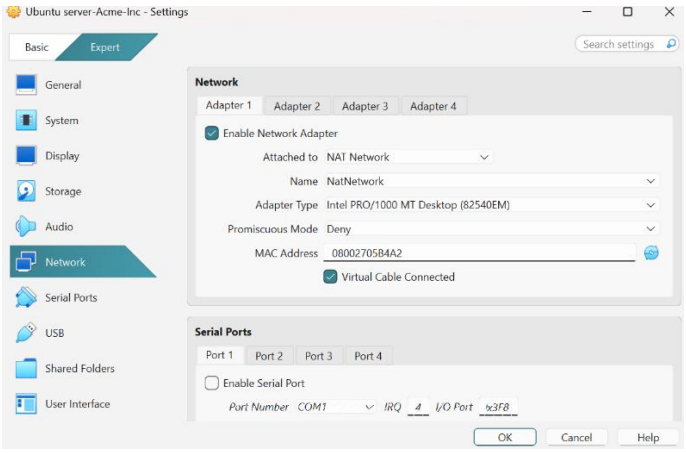
NAT Network Created on VirtualBox



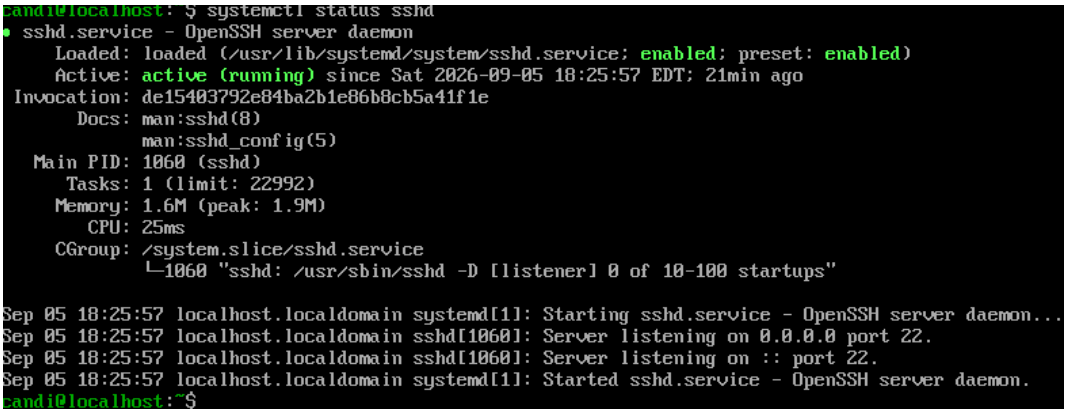
CentOS - NAT Network connection



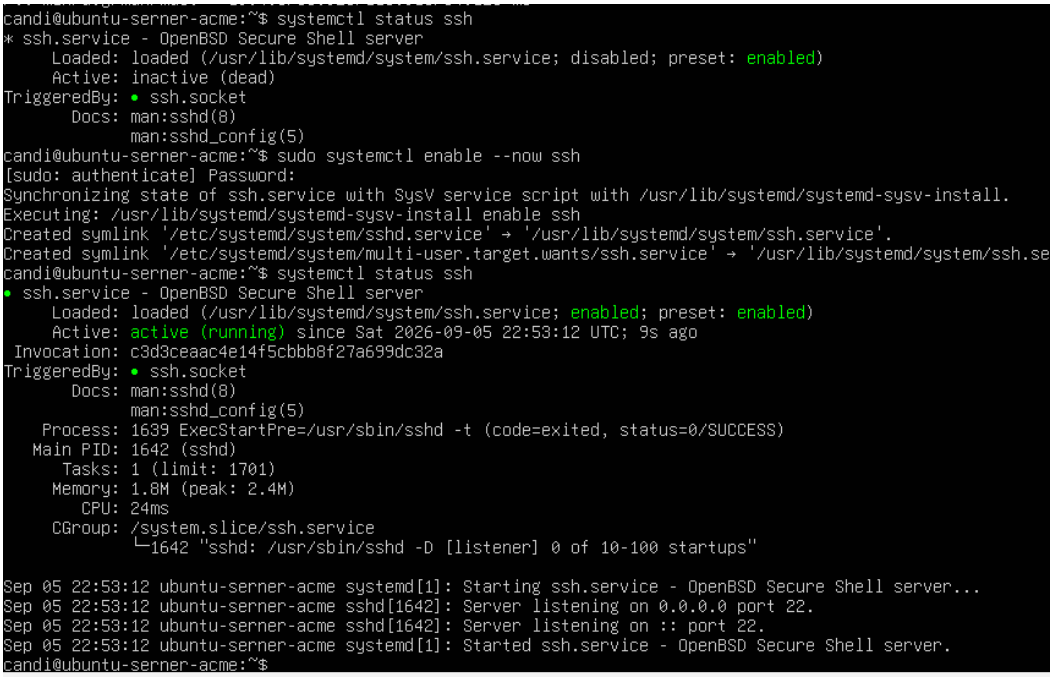
Ubuntu - NAT Network connection



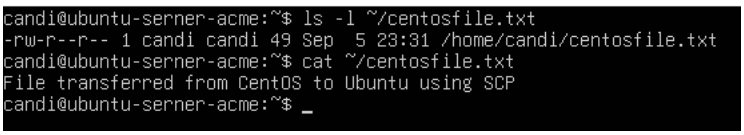
CentOS - SSH Running



Ubuntu - SSH running



Ubuntu successfully received CentOSfile.txt from the CentOS server using SCP



CentOS successfully received testfile.txt from the Ubuntu server using SCP



Sources:

https://www.aholdengouveia.name/LinuxAdmin/Documentation.html

https://ubuntu.com/server/docs/

https://manpages.ubuntu.com/manpages/focal/man8/ping.8.html

https://ubuntu.com/server/docs/reference/cli-cheatsheet/

https://docs.oracle.com/en/virtualization/virtualbox/7.2/user/networkingdetails.html

https://docs.oracle.com/en/virtualization/virtualbox/7.2/user/index.html

scp(1) - https://access.redhat.com/sites/default/files/attachments/12052018_systemd_6.pdf OpenBSD manual pages

Red Hat Enterprise L https://access.redhat.com/sites/default/files/attachments/12052018_systemd_6.pdf inux cheat sheets for system administrators - Red Hat Customer Portal

12052018_systemd_6 https://access.redhat.com/sites/default/files/attachments/12052018_systemd_6.pdf.pdf